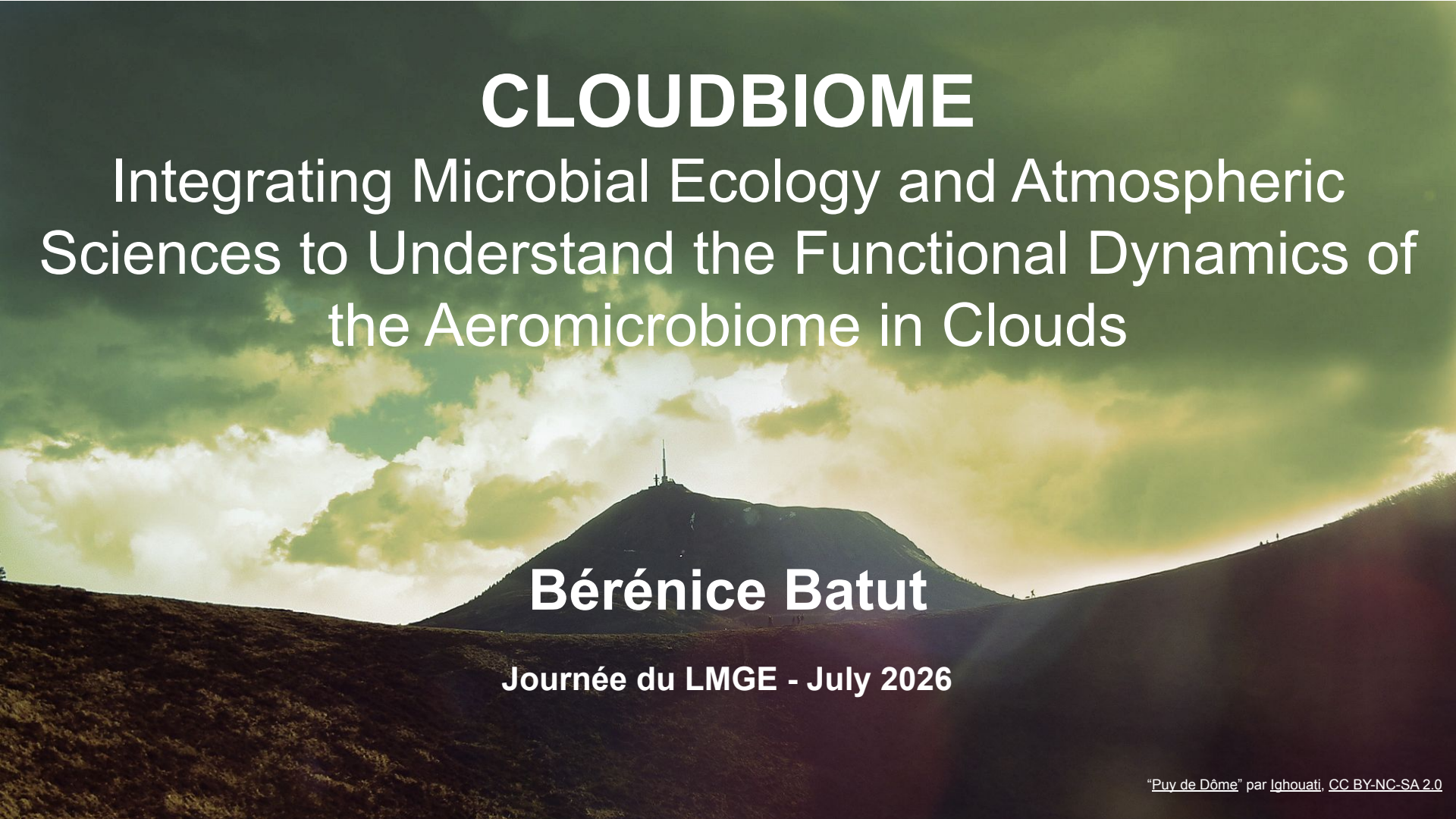
CLOUDBIOME

Integrating Microbial Ecology and Atmospheric Sciences to Understand the Functional Dynamics of the Aeromicrobiome in Clouds

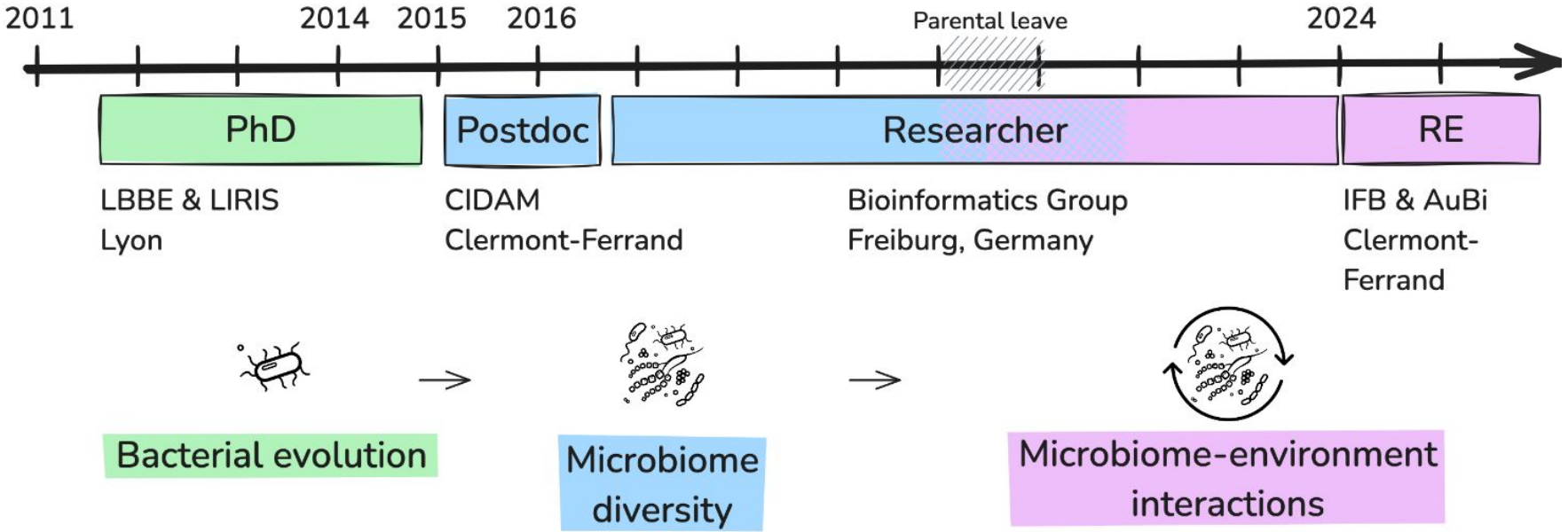
Bérénice Batut

Journée du LMGE - July 2026



"Puy de Dôme" par Ighouati, CC BY-NC-SA 2.0

From bacterial evolution to understanding microbial ecosystems through bioinformatics



environment by HideMaru, microbiome by Stefanie Pesche from Noun Project (CC BY 3.0)

Galaxy

A robust open-source data analysis platform, with an active international community

Platform focused on **reproducible, transparent & FAIR research since 2005**

A **powerful workflow manager** to build and run **workflows**

20k+ Citations	**10k+** Tools	**700+** Workflows	**~250** Contributors/year	**100+** Servers	**650k+** Users

The Galaxy Community, NAR, 2018, 2022, 2024, 2026.

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500+ Tutorials

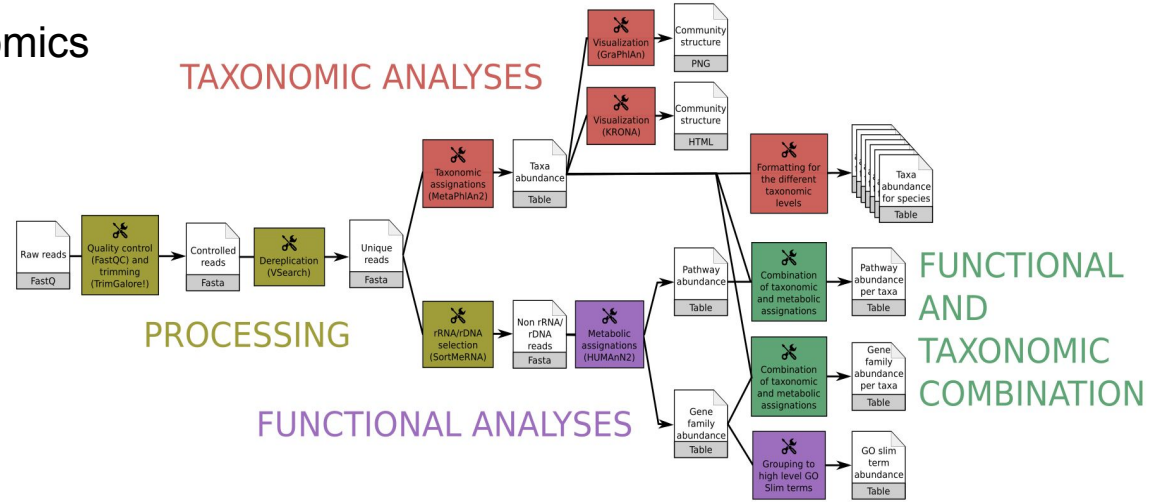


Sustainable, global and widely-used platform for accessible training

The Galaxy Community, NAR, 2018, 2022, 2024, 2026. Hiltemann et al, PLOS Computational Biology, 2023.

Methodologies for FAIR Microbiome-Environment Interaction Research

Metagenomics / Metatranscriptomics

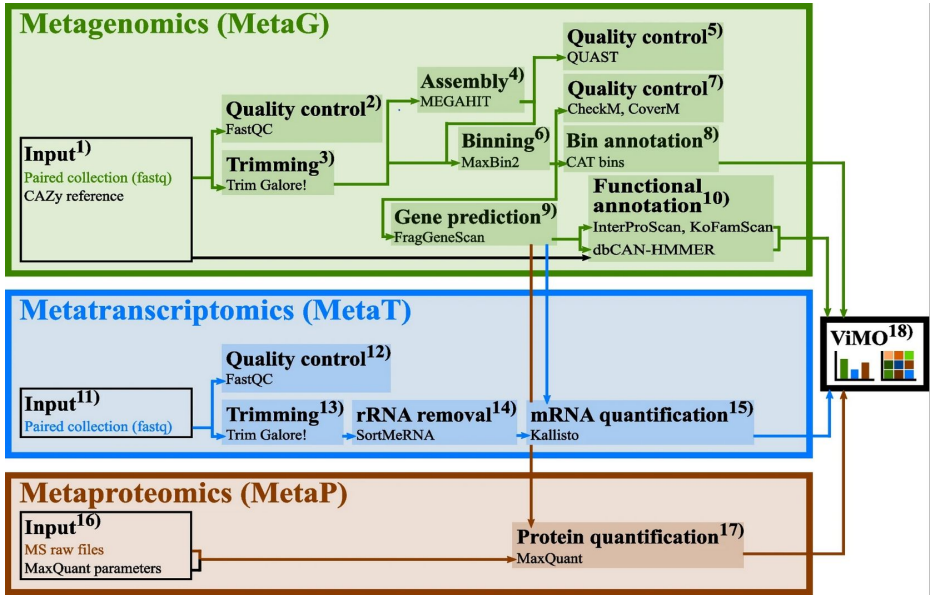


Batut et al, Gigascience, 2018. Mehta et al, F1000, 2021

Methodologies for FAIR Microbiome-Environment Interaction Research

Metagenomics / Metatranscriptomics

Integrative meta-omics



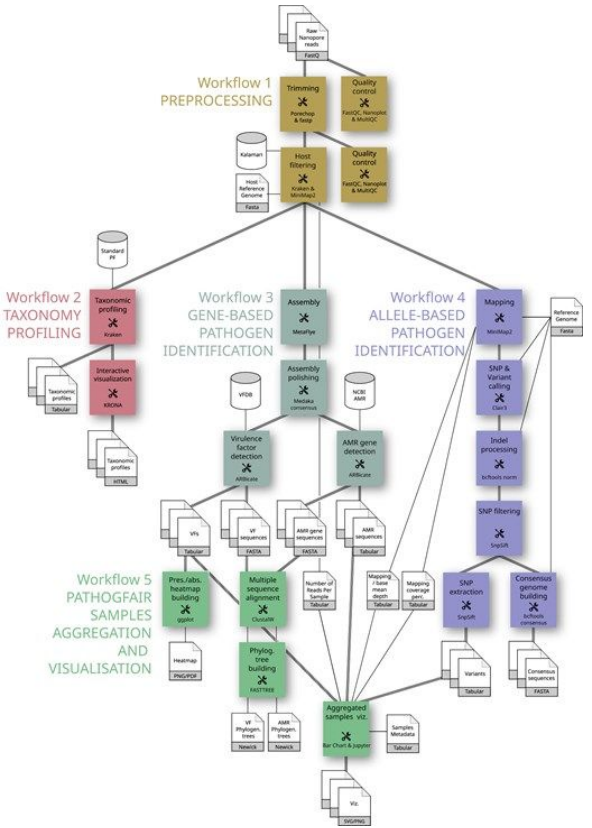
Schiml et al, Environmental Microbiome, 2023.

Methodologies for FAIR Microbiome-Environment Interaction Research

Metagenomics / Metatranscriptomics

Integrative meta-omics

Pathogen detection



Nasr et al, Gigascience, 2025.

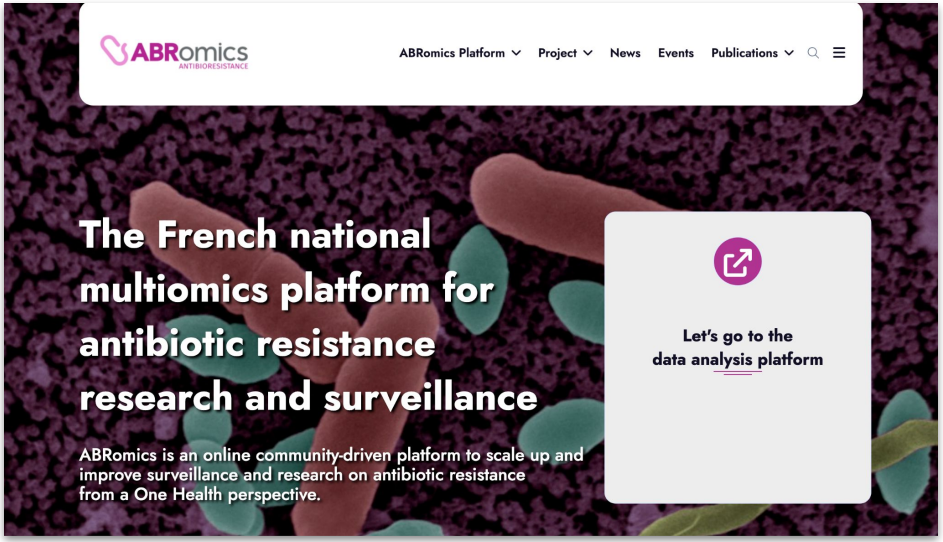
Methodologies for FAIR Microbiome-Environment Interaction Research

Metagenomics / Metatranscriptomics

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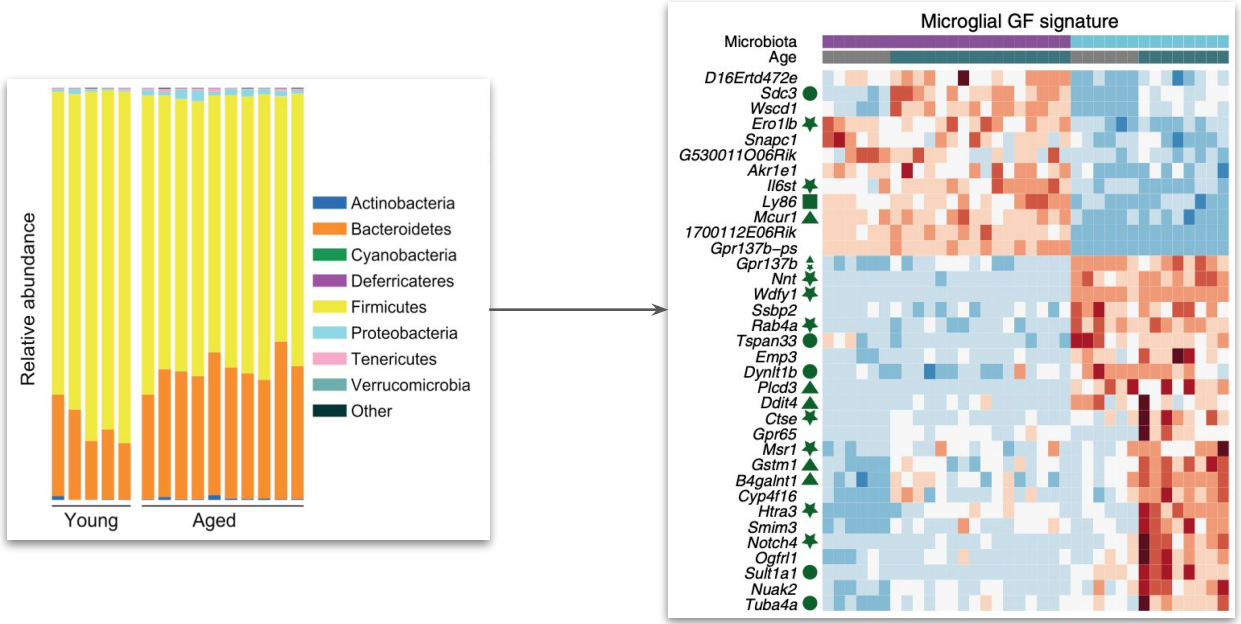
Antibiotic resistance



Siguret et al, in preparation.

Studies of Microbiome-Environment Interactions

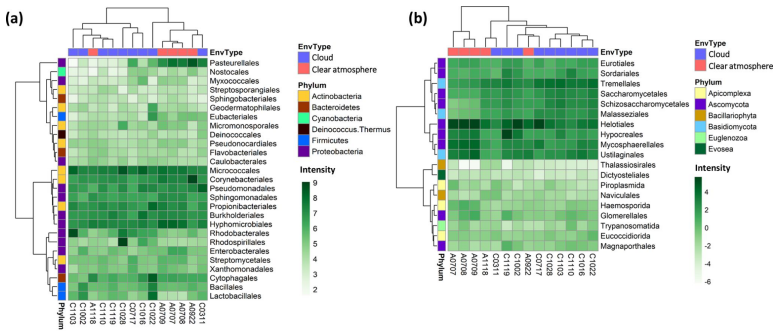
🐭 Impact of gut microbiota on microglial cell aging

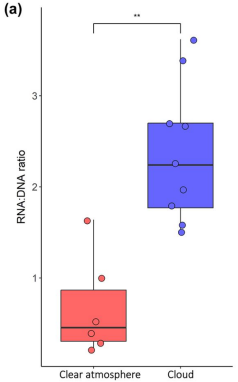


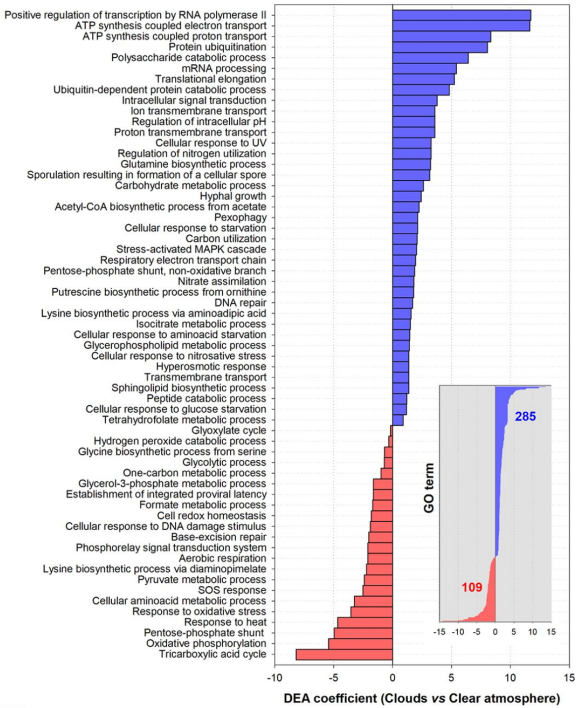
Mossad, Batut, et al., Nat Neurosc, 2022.

Studies of Microbiome-Environment Interactions

- 🐭 Impact of gut microbiota on microglial cell aging
- ☁️ Influence of clouds on airborne microorganisms

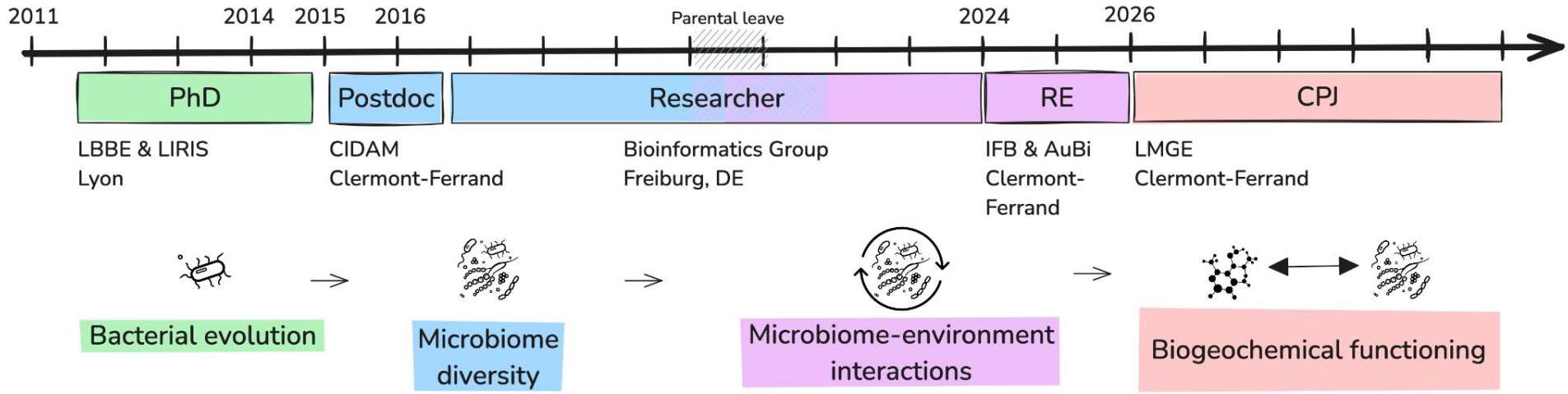






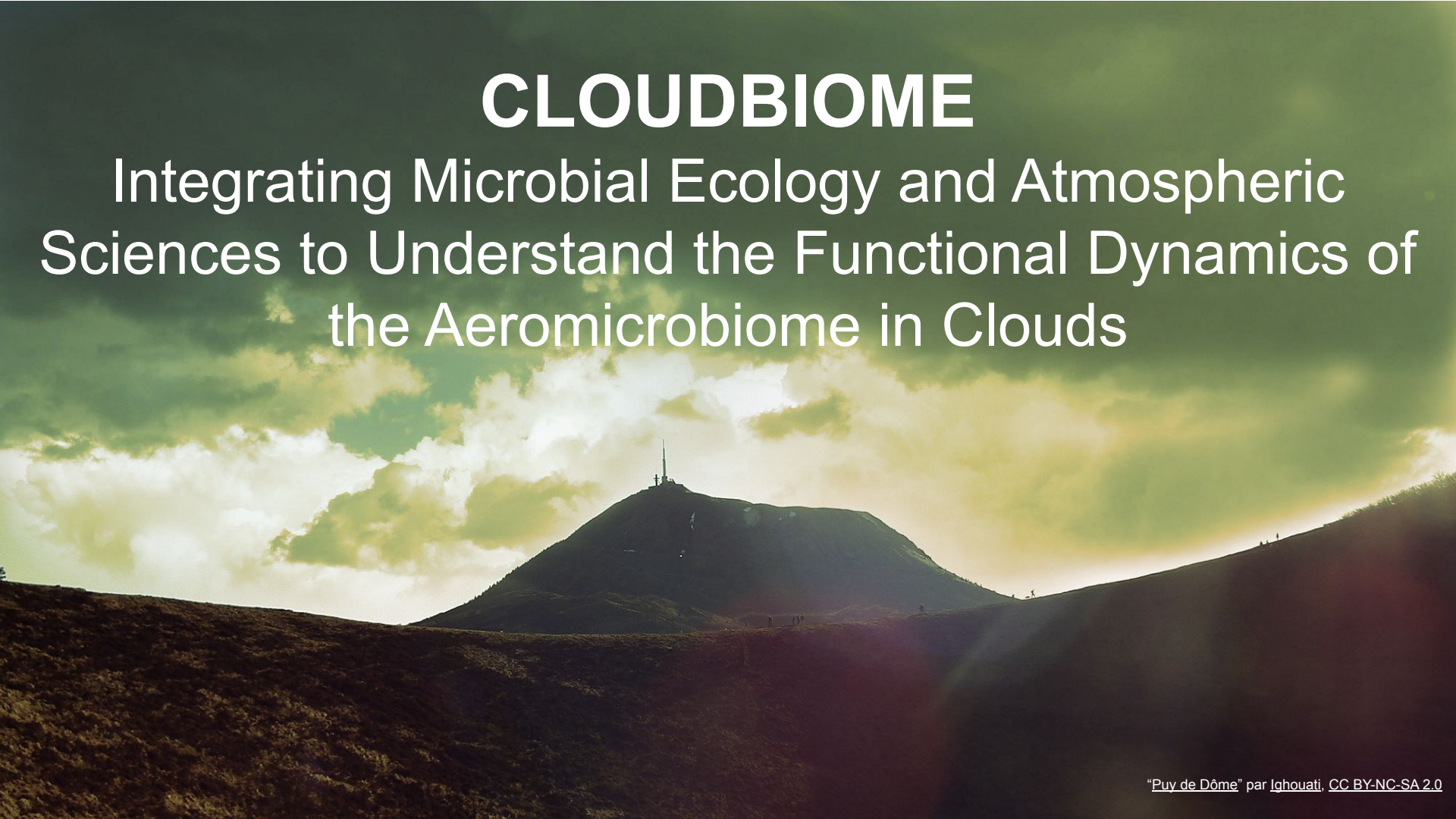
Péguilhan, et al., Biogeosciences, 2025.

Next step


2011
2014
2015
2016
Parental leave
2024
2026
PhD
Postdoc
Researcher
RE
CPJ
LBBE & LIRIS
Lyon
CIDAM
Clermont-Ferrand
Bioinformatics Group
Freiburg, DE
IFB & AuBi
Clermont-
Ferrand
LMGE
Clermont-Ferrand
Bacterial evolution
Microbiome
diversity
Microbiome-environment
interactions
Biogeochemical functioning

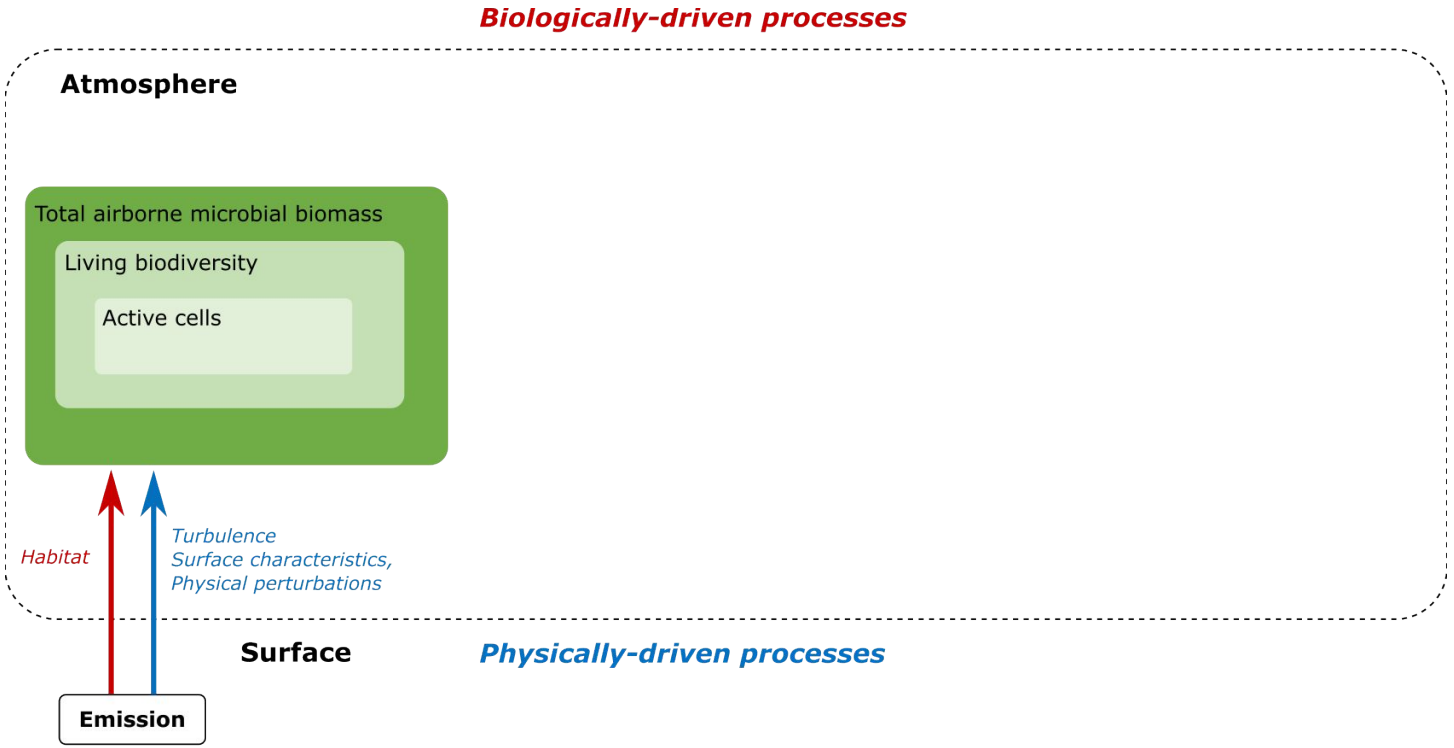
CLOUDBIOME

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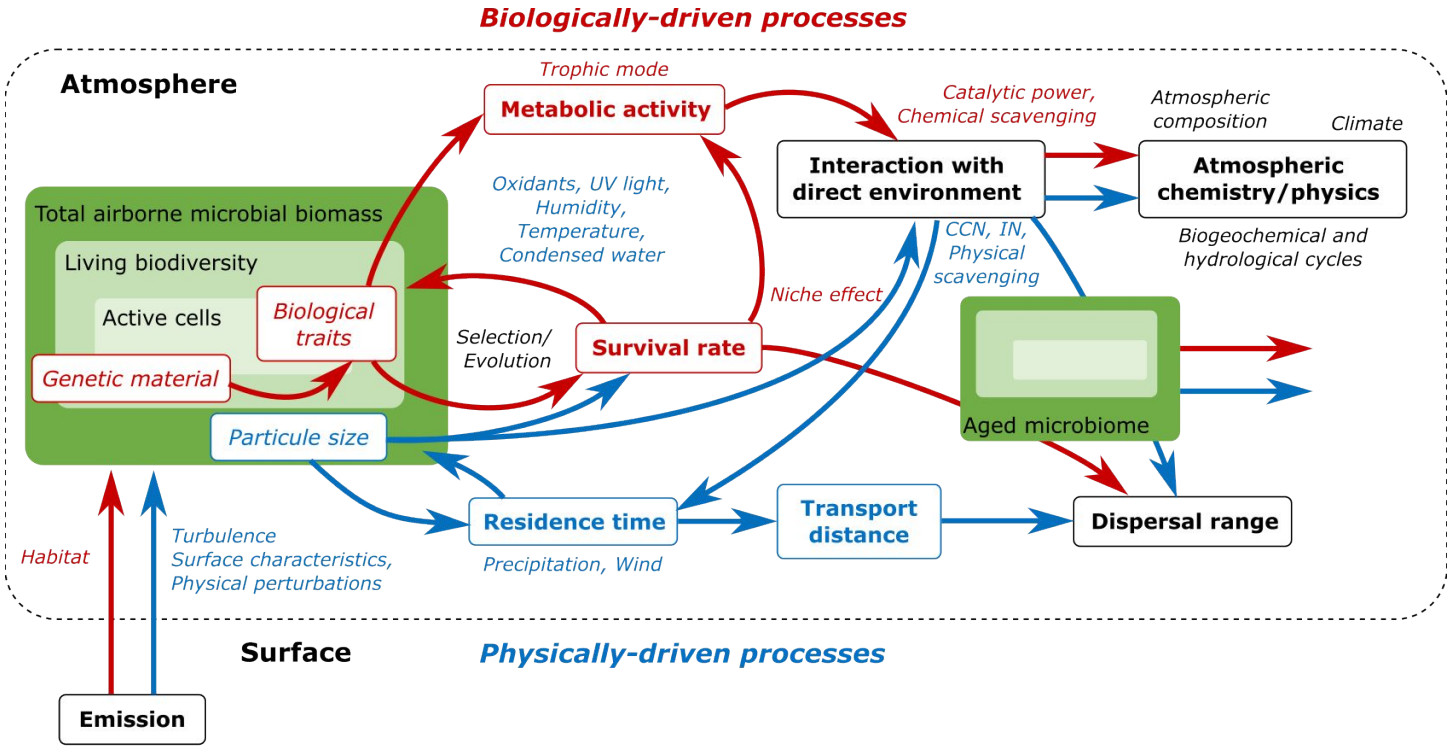
"Puy de Dôme" par Ighouati, CC BY-NC-SA 2.0

Aeromicrobiome



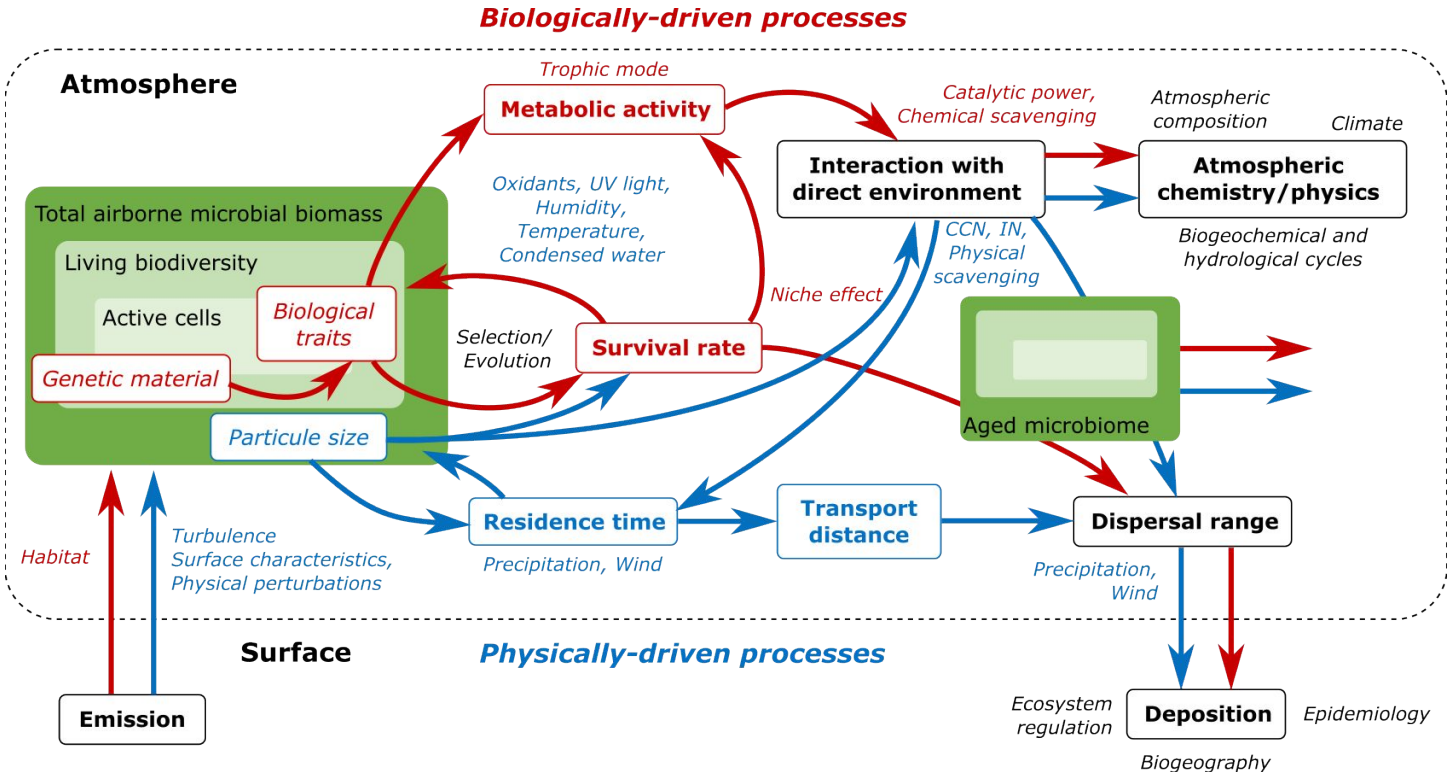
Adapted from [Amato *et al.*, 2023]

Aeromicrobiome



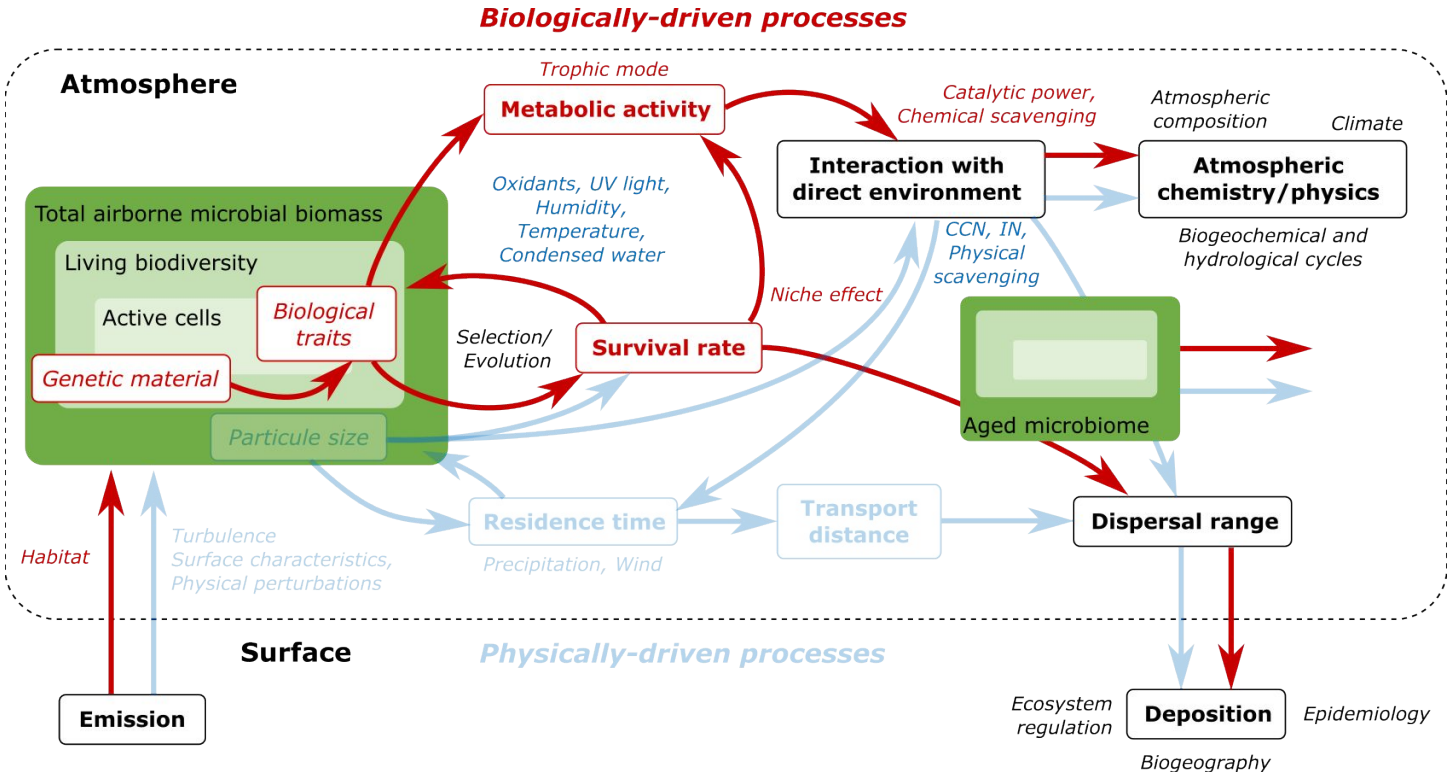
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Aeromicrobiome



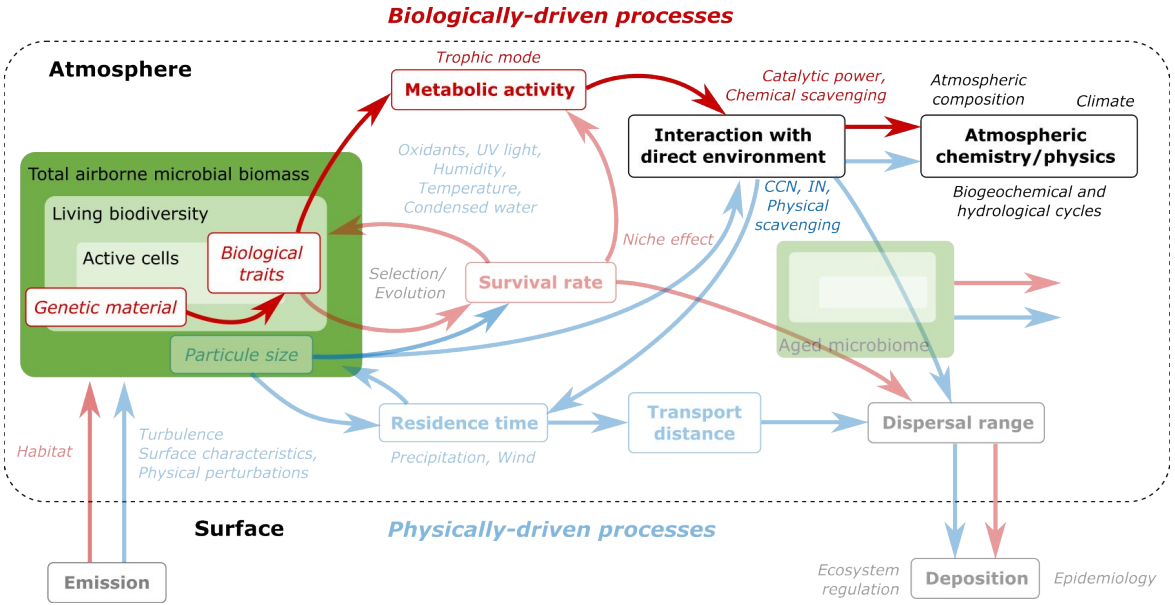
Adapted from [Amato *et al.*, 2023]

Focus on the biologically-driven processes



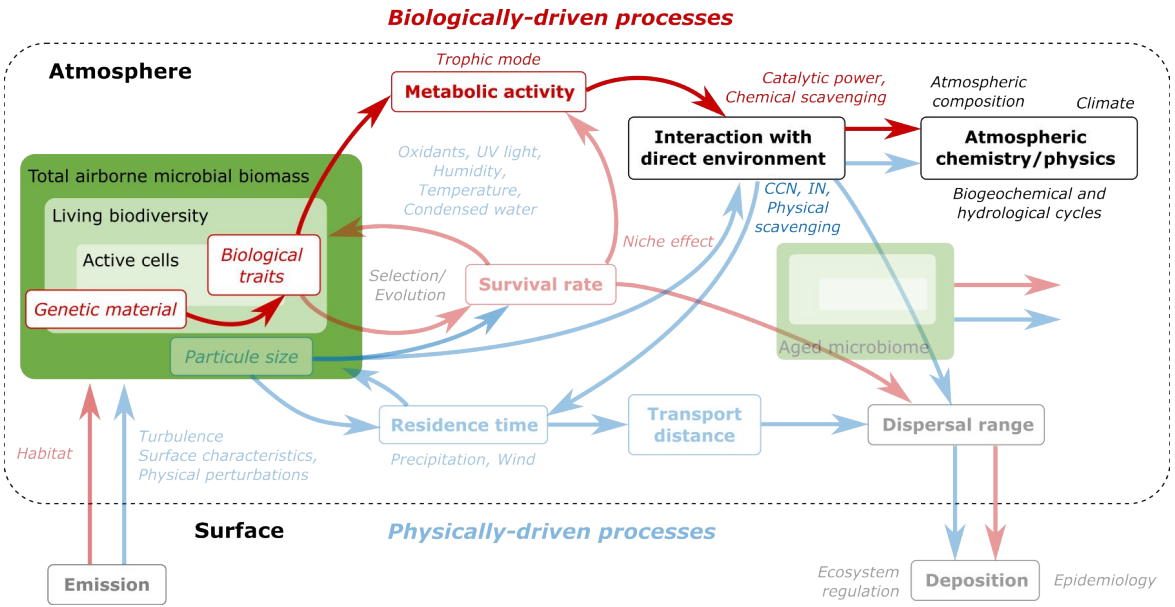
Adapted from [Amato *et al.*, 2023]

Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



Adapted from [Amato *et al.*, 2023]

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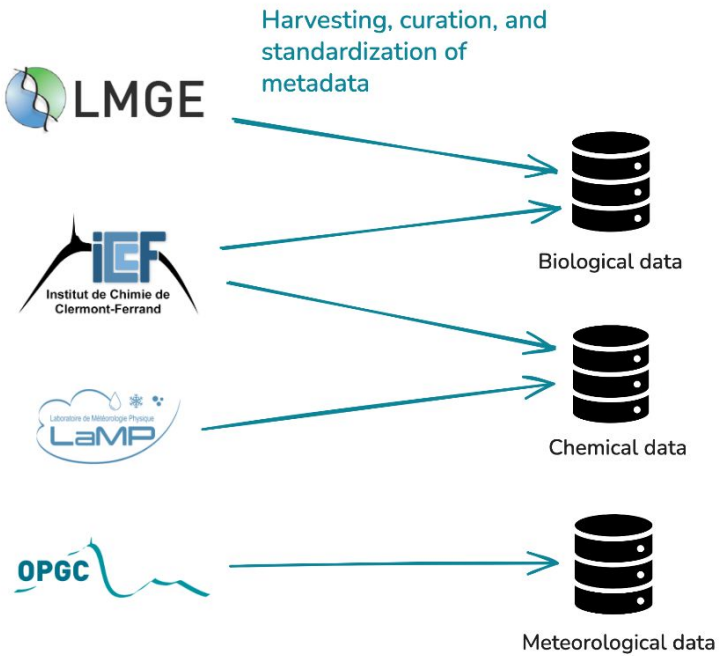


Some scientific questions

- How do **cloud type**, **duration**, and **environmental factors** shape **microbial diversity**, **composition**, and **transcriptional activity** in **bioaerosols** at **Puy de Dôme**?
- Do **active cloud microorganisms** drive **atmospheric processes** (e.g., **carbon cycling**, **aerosol formation**) through **enriched metabolic pathways**, and how do these influence **cloud chemistry**?

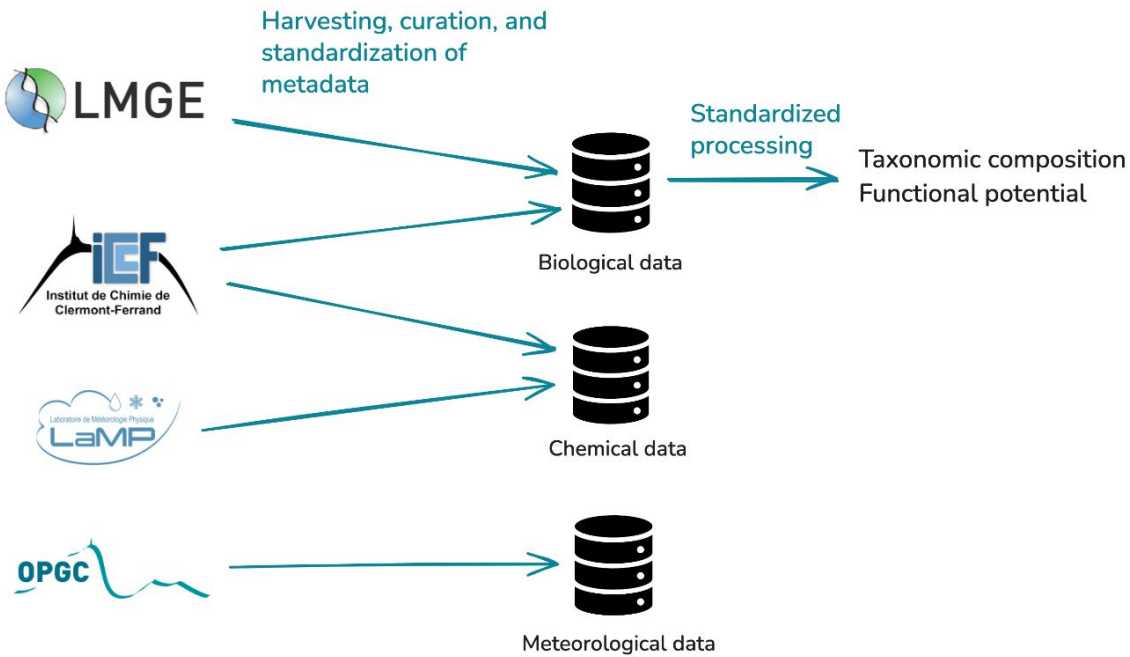
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Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



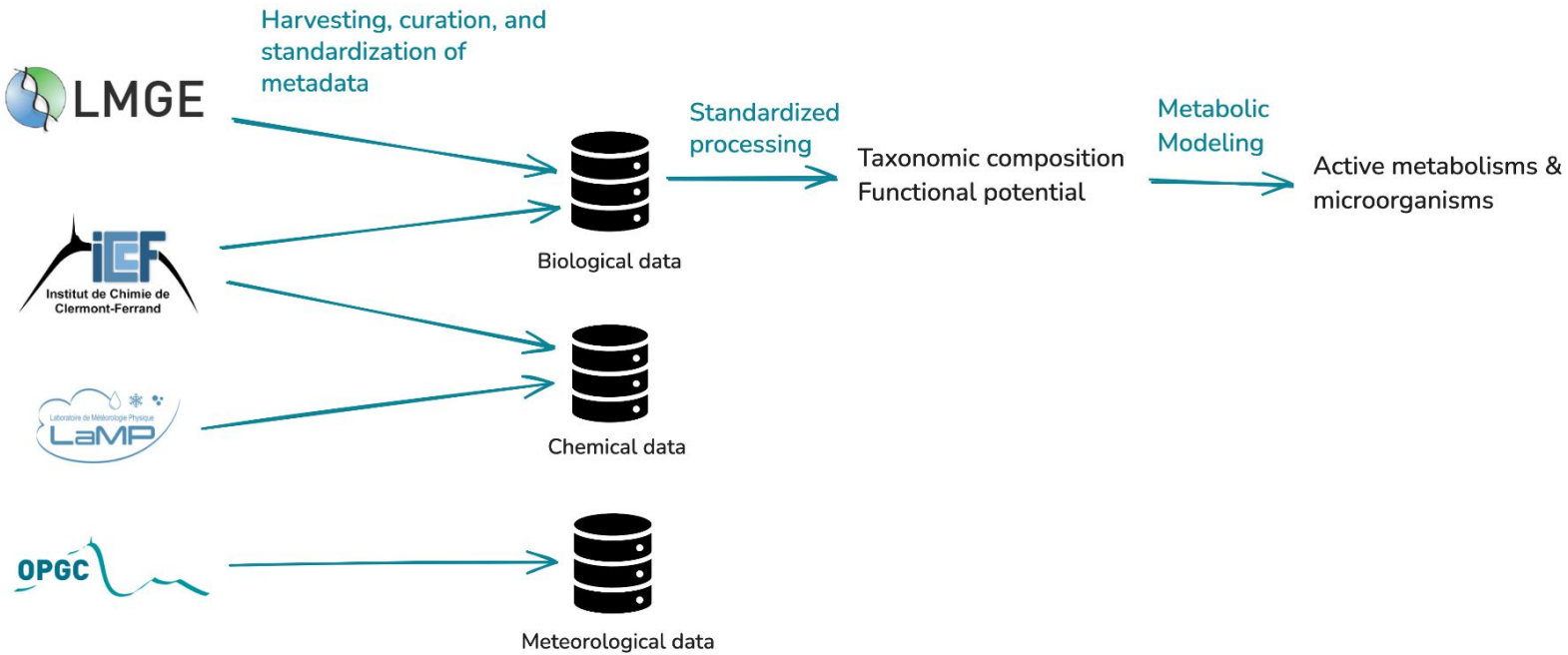
database by Lewen Design from Noun Project (CC BY 3.0)

Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



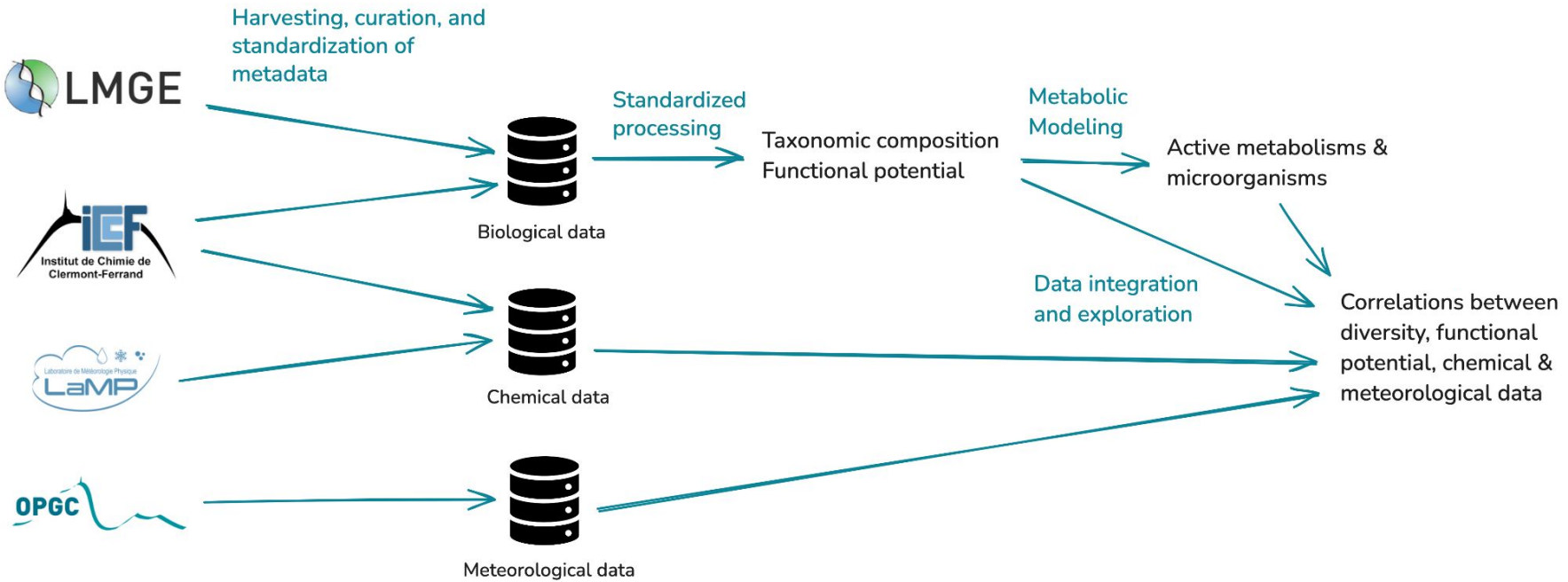
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Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



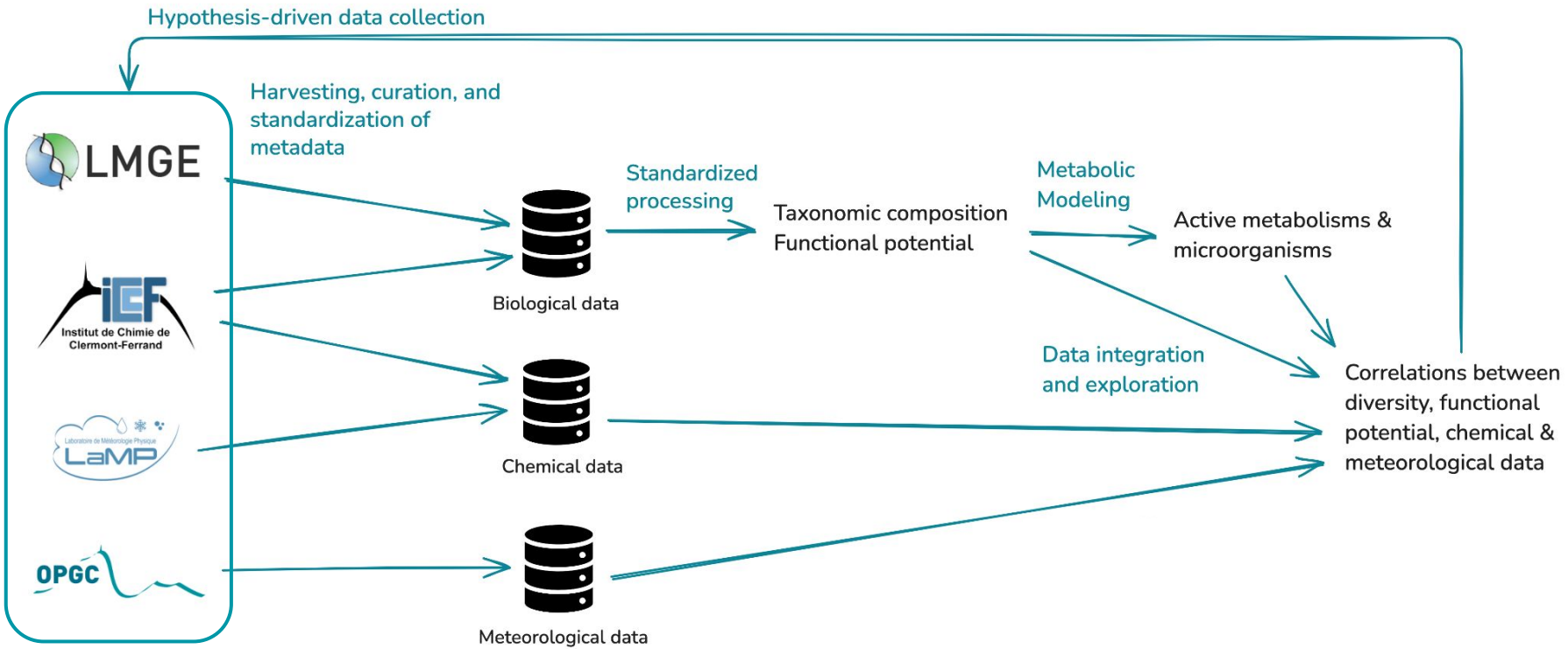
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Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



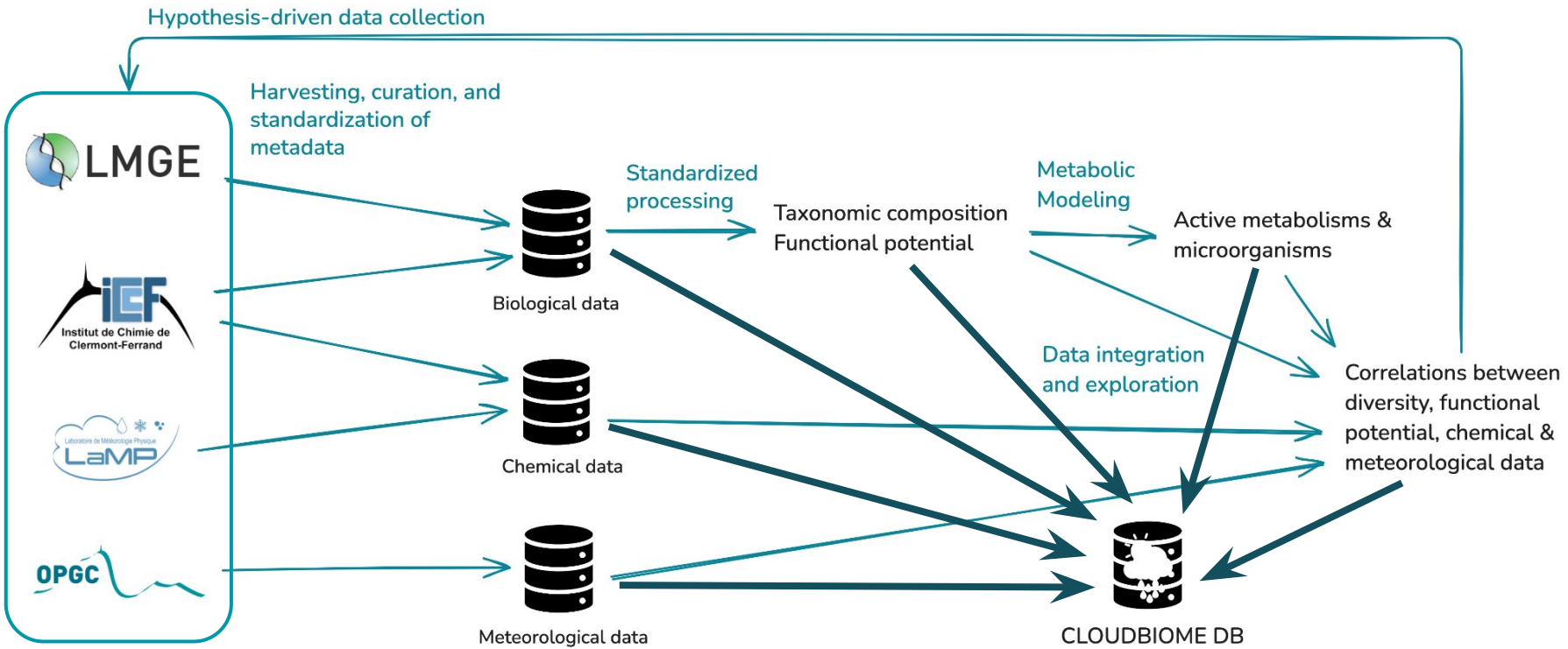
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Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



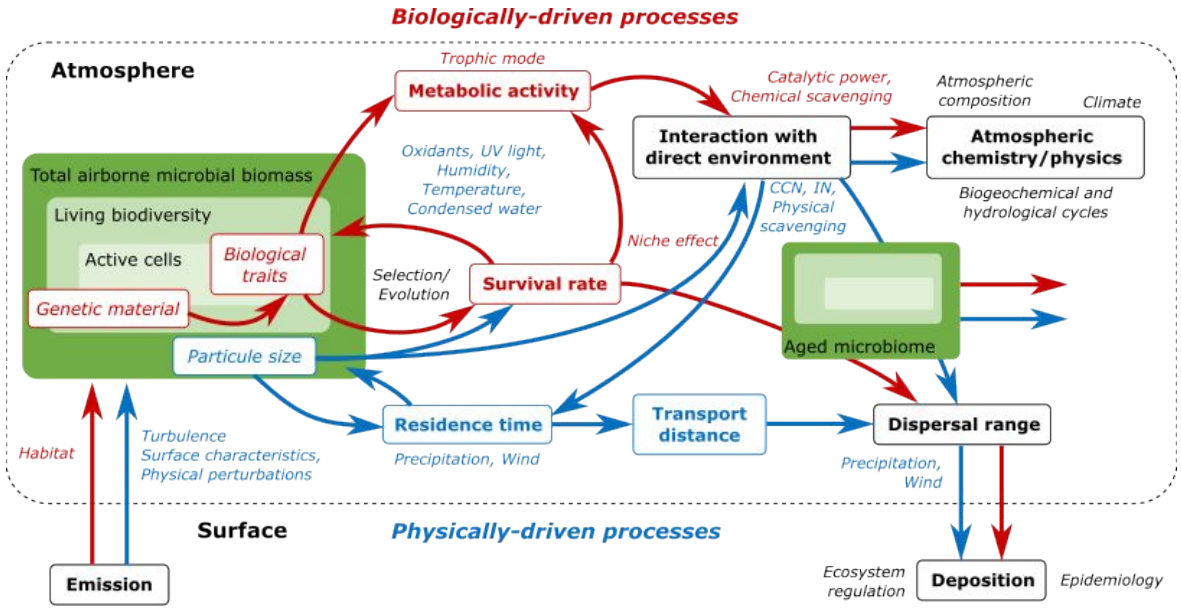
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Theme 1: Bioaerosol Functional Diversity and Its Role in Cloud Biogeochemistry



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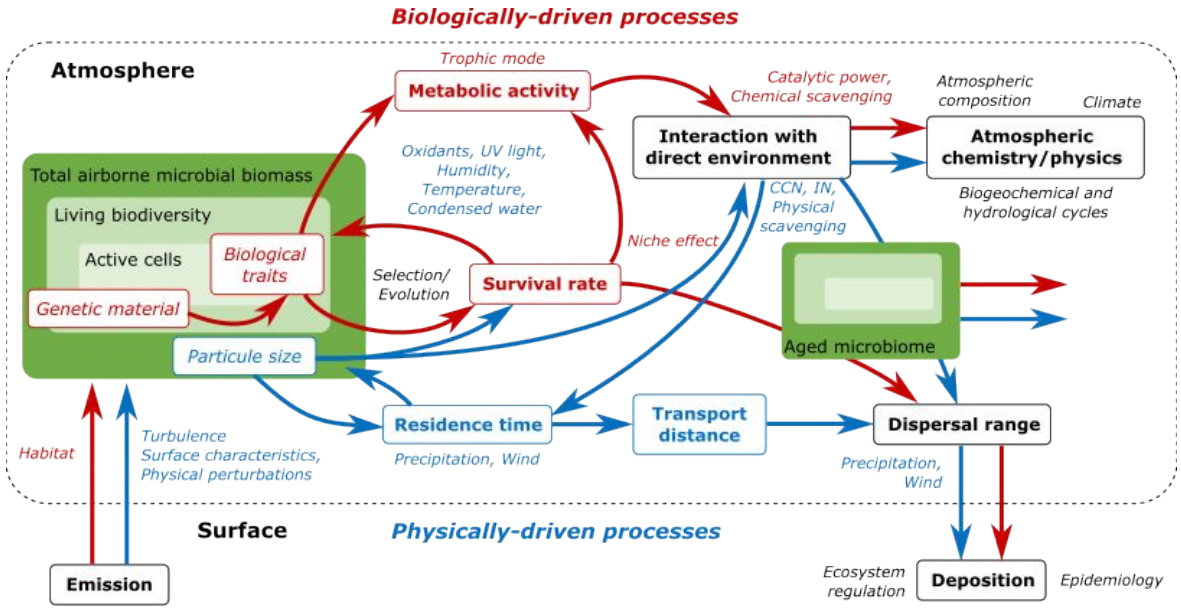
Understanding the Functional Dynamics of the Aeromicrobiome in Clouds



Theme 1: **Bioaerosol Functional Diversity** and Its **Role** in **Cloud Biogeochemistry**

Adapted from [Amato *et al.*, 2023]

Understanding the Functional Dynamics of the Aeromicrobiome in Clouds

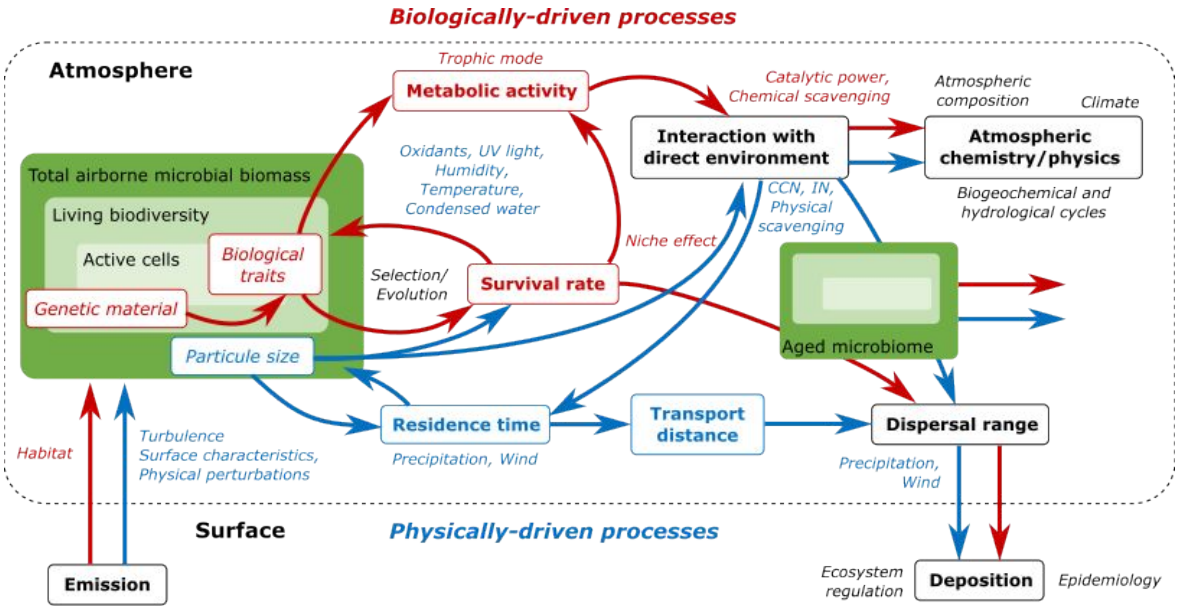


Theme 1: **Bioaerosol Functional Diversity** and Its **Role** in **Cloud Biogeochemistry**

Theme 2: **Adaptations** of Microorganisms to **Atmospheric Transport**

Adapted from [Amato *et al.*, 2023]

Understanding the Functional Dynamics of the Aeromicrobiome in Clouds



Theme 1: **Bioaerosol Functional Diversity** and Its **Role** in **Cloud Biogeochemistry**

Theme 2: **Adaptations** of Microorganisms to **Atmospheric Transport**

Theme 3: **Transfer** of Biological Materials Through the **Water Cycle**

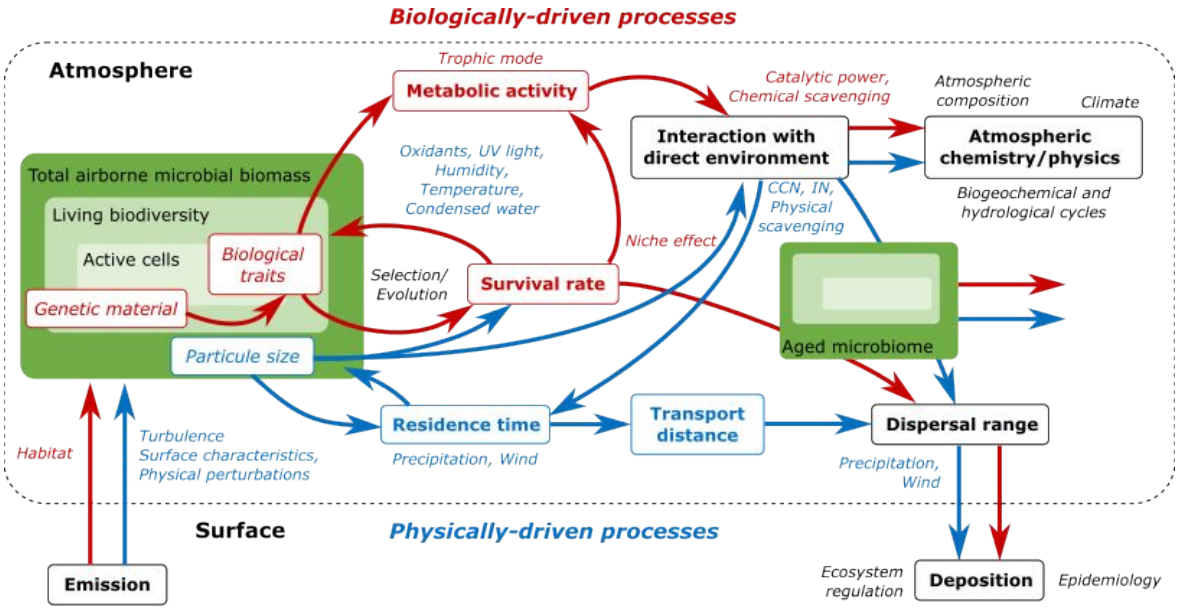
Adapted from [Amato *et al.*, 2023]

Thank You!



“Puy de Dôme” par Ighouati, CC BY-NC-SA 2.0

Understanding the Functional Dynamics of the Aeromicrobiome in Clouds



Theme 1: **Bioaerosol Functional Diversity** and Its **Role** in **Cloud Biogeochemistry**

Theme 2: **Adaptations** of Microorganisms to **Atmospheric Transport**

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